

Perspectives on Visual Analytics for Advancing Imaginative Intelligence

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Abstract

With recent developments in multimodal large language models (MLLMs), the capability in cross-modality authoring enables imaginative intelligence. However, such advancements remain constrained by the inadequacy of reliability due to unexplained mechanisms and the hallucination of artificial intelligence. To address this issue, we summarize progress on imaginative intelligence by surveying MLLM techniques. Aimed at fully activating imaginative intelligence, we identify seven key prescription requirements to steer MLLMs and propose a visual prescription workflow that inherently satisfies the identified requirements. Existing research has proven that visual analysis can enable effective and efficient communication between users and complex mechanisms. Therefore, we summarize open research opportunities for visual analytics based on the workflow and describe corresponding practical possibilities.

Keywords: Imaginative intelligence, multimodal large language models, visual prescription, visual analytics

1. Introduction

Multimodal large language models (MLLMs) [1, 2] integrate diffusion methodologies, composite architectural frameworks, and additional technological innovations to synthesize visual content from cross-modal inputs (e.g., textual or pictorial). Their capacity to generate novel and semantically coherent multimodal outputs, including text-to-video (T2V) or image-to-video (I2V) content, depends on sophisticated processes of inference, association, and generative capability that extend beyond the literal interpretation of prompts. This cognitive function, characterized as imaginative intelligence, works mainly by visualizing information (e.g., translating abstract conceptualizations into visual representations). This embodied cognitive capability facilitates transformative applications across domains, including creative design (advertising, interactive entertainment), educational contexts (experiential learning), and human-AI collaborative environments

(conceptual development, intelligent assistance systems).

Imaginative intelligence demonstrates the capacity to render incomplete descriptions with visual fidelity, complex dynamics, and narrative coherence through inferential detail generation [3, 4, 5]. The evolution of this intelligence has transcended basic information transcription to now support or even lead in domains such as artistic conceptualization, proposal development, and speculative scenario exploration [6]. Nevertheless, despite ongoing model refinement, hallucinations—manifested as factual fabrications or prompt-inconsistent content generation—persist and are considered intrinsic byproducts of contemporary generative methodologies [7]. These limitations have engendered reliability concerns and constitute a significant impediment to broader implementation in critical domains such as healthcare consultation and autonomous vehicle systems, where precision and safety are essential [7]. Within these contexts, imaginative intelligence cannot ensure output validity, nor can it assess or communicate uncertainties stemming from hallucinations [7]. Concur-

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rently, users increasingly require precise generative control, granular output correction capabilities, and instantaneous iterative feedback [8]—requirements that current mainstream opaque generative approaches inadequately address.

The widespread implementation of imaginative intelligence necessitates addressing reliability challenges through a systematic framework for analyzing AI-generated content, identifying inaccuracies, and informing iterative model refinement. While text-based models have seen considerable advancements in this domain, video generation technology has progressed more gradually due to inherent complexities in its multimodal and temporal dimensions [8]. Besides, current automatic quantitative evaluation methodologies demonstrate limitations in assessing sophisticated semantic coherence and physical plausibility, whereas manual evaluation protocols become prohibitively resource-intensive as video duration and resolution increase. This context requires an analytical approach that integrates human discernment with computational efficiency, capable of both error detection and performance enhancement. We propose visual prescription as this methodological solution: a procedural framework leveraging effective human-AI collaboration and feedback mechanisms to detect various anomalies, ranging from hallucinations to physical implausibility, in generated video content and systematically enhance model performance through iterative development.

After research, we found that both AI systems and humans face three key challenges in visual prescription. First, videos inherently contain an enormous density of information, spanning verification dimensions from pixel information to high-level semantic coherence. Second, the verification process must adhere to multiple criteria, many of which, such as laws of physics, common sense reasoning, and temporal consistency, are implicit constraints rather than explicit prompts; yet these default constraints are critical for real-world scenarios (e.g., generating videos

that obey physical laws even when they are not specified) [9]. Third, factors such as perspective occlusion, rapid motion blur, information overload, and human perceptual limitations make it difficult for experts to efficiently make completely accurate and comprehensive judgments on all aspects of video content, even with the assistance of automated tools.

To address these challenges, we employ visual analytics (VA) as the foundational technology for implementing visual prescription. Generative video represents a data modality characterized by both informational density and temporal complexity, wherein VA enhances analytical reasoning through the integration of intuitive graphical interfaces, interactive methodologies, and computational algorithms, offering significant advantages in processing multifaceted, high-dimensional, and temporal data [1]. This investigation examines how VA can facilitate visual prescriptions for imaginative intelligence and evaluates appropriate analytical approaches to meet these requirements. Our research extends existing research on interactive feedback and alignment for MLLMs, while accounting for the distinctive characteristics of video generation, and aims to establish a conceptual framework that motivates the future design and development of related analytical systems for imaginative intelligence.

In this research, we present two principal contributions:

- We synthesize the current advancements and developments in imaginative intelligence.
- We introduce an integrated approach that combines prompt-driven video authoring with VA-based error detection, user-guided correction, and precise iteration.
- We identify research directions at the nexus of generative AI and VA, encompassing interactive error detection mechanisms, multimodal prompt engineering, and provenance visualization, while delineating some practice possibilities to test the efficacy of our pro-

posed approach.

2. Progress of Imaginative Intelligence

In this section, we introduce the progress of imagination intelligence and identify key challenges in translating creative intent to visual outputs. To address these challenges, we review existing interaction methods developed for LLMs and discuss the gap for MLLMs. Based on identified gaps, we formulate requirements for visual prescription systems tailored to advanced video generation models.

2.1. Progress of Imaginative Intelligence

Imaginative intelligence can lead to groundbreaking creations or inappropriate responses due to AI hallucinations [7]. In recent years, driven by emerging generative paradigms, imaginative intelligence has made significant advances in video generation, with new methods and systems proposed [6, 10]. In particular, Sora has been widely cited as a representative system for its high-fidelity video and clip generation [3]; Lumiere achieves novel semantic alignment through multimodal attention fusion [5]; and Veo 2 enhances motion continuity and scene consistency through large-scale pre-trained encoders and controllable frame interpolation [4]. Next, we briefly introduce the framework of these three SOTA video generation techniques.

The current landscape of video generation is characterized by the coexistence of three primary architectural paradigms [10], each of which presents unique trade-offs and is represented by prominent models: Diffusion Transformer (DiT) exemplified by OpenAI’s Sora [3, 11], Latent Diffusion Model (LDM) represented by Google’s Veo 2 [4, 12], and Specialized Spatio-Temporal U-Net (STUNet) showcased by Google’s Lumiere [5]. Fundamentally, diffusion models form the basic generative process [13], with the core distinction being in the backbone architecture (Transformer vs. U-Net variant) or the oper-

ational space (pixel/patch vs. latent space). For clarity, we discuss these as three different approaches below:

DiT: The DiT framework, exemplified by Sora, incorporates Transformer architectures within the diffusion paradigm. By conceptualizing video as sequences of spatio-temporal patches, this approach leverages the fundamental capacity of Transformers to model extensive contextual dependencies [14], thereby facilitating the representation of sophisticated temporal dynamics and narrative structures. This methodology demonstrates considerable potential for scalability [11], enabling systems to capitalize on enhanced data volumes and computational resources. Nevertheless, DiT architectures necessitate substantial computational infrastructure, may encounter training stability issues, and occasionally manifest limitations in the accurate simulation of complex physical phenomena or in maintaining consistent object representation across frames, occasionally resulting in localized structural anomalies [3].

LDM: LDM exemplified by Veo 2, prioritizes computational efficiency [12]. This paradigm employs a variational encoder (VAE) to compress high-dimensional video data into a lower-dimensional latent space, where the diffusion process subsequently occurs [12]. This dramatically reduces computational costs compared to pixel-space diffusion, facilitating more efficient generation, particularly for high-resolution outputs. LDMs also offer flexibility in the integration of conditioning signals. The primary limitation lies in the dependence on the VAE’s reconstruction quality; information loss during compression can affect fine details in the final video [12]. Furthermore, the learned latent space may not be easily interpretable or controllable [15], and errors can potentially accumulate throughout the encoding-diffusion-decoding pipeline, sometimes impacting the precise adherence to complex prompts.

STUNet: Presented by Lumiere. It adapts the conventional U-net architecture by explicitly incorporating temporal modelling, often through convolutions. A key fea-

ture of Lumiere in particular is the design to generate the entire video duration in a single forward pass [5], which prioritizes global temporal coherence, resulting in exceptionally smooth motion dynamics, which can be beneficial for subsequent editing tasks. The main drawback is the significant computational and storage requirements associated with processing the entire duration simultaneously, which often limits the achievable video length or requires generation at lower resolutions followed by upscaling [5]. Handling scenes with extremely high detail complexity or frequent cuts can also be challenging for this single-pass approach.

Comparative analysis of architectures: In the three video generation paradigms, DiT embeds Transformer blocks into the diffusion denoising process to exploit global self-attention, leading to significant improvements in visual fidelity and narrative coherence; however, DiT’s reliance on large-scale model and data parallelism imposes significant computational and storage costs, often leading to optimization instability during training, and can still introduce small local biases when modeling complex physical interactions or object consistency [14, 3, 11]. LDM, instead, compresses high-dimensional video into a lower-dimensional VAE latent space where diffusion occurs, drastically reducing computational requirements and enabling efficient high-resolution generation. Such flexibility supports rich cross-attention conditioning, but suffers from loss of detail due to VAE reconstruction limitations, opaque latent representations, and the potential for error accumulation throughout the encode-diffuse-decode pipeline [12, 15]. Finally, STUNet explicitly integrates temporal convolutions or attention into a U-net backbone to generate entire video segments in a single forward pass, ensuring exceptional motion smoothness and global temporal consistency, which benefits downstream editing, but at the cost of large memory footprints, constraints on video duration or resolution, and reduced adaptability to highly complex scenes or rapid shot changes [5].

In common, these architectures are extremely complex and their processing is difficult to understand, making it difficult to locate the errors. While VA offers a complementary solution space by coupling intuitive graphical summaries, interactive error highlighting, and automated analysis pipelines to expose physical plausibility, semantic misalignment, and hallucinations in generated video; such VA tools can guide users through multi-dimensional diagnostics and provide the structured feedback loops necessary for our VA-driven visual prescription framework, laying the groundwork for systematic model debugging and iterative enhancement [16, 17].

2.2. Video Generation Evaluation with examples

Evaluating the quality of generated video is crucial in VA and is significantly more challenging than assessing static images, primarily due to the introduction of the temporal dimension, which requires the assessment of dynamic consistency, motion realism, and long-term coherence. Traditional image-based metrics such as PSNR and SSIMs [18], while easy to compute, are fundamentally inadequate for this task as they operate on a frame-by-frame basis, largely ignoring the crucial temporal dynamics and often correlating poorly with human perceptual judgment [19]. This shortcoming spurred the development of video-centric evaluation methods. Early dynamic methods, such as Fréchet Video Distance (FVD) [19], which assesses the distributional similarity between real and generated video features (often extracted via models such as I3D) [20], represented an improvement. However, these dynamic metrics were subsequently proven to have sensitivity to feature extractor biases (e.g., I3D’s potential content bias and insensitivity to temporal distortions), significant data requirements for stable estimation, and sometimes lagging behind the rapid advances in generative models [21]. The current SOTA evaluation paradigm has converged on comprehensive, multi-dimensional benchmarks designed to provide a more holistic and fine-grained

evaluation. Recognizing that video quality is not monolithic, frameworks such as VBench [21], EvalCrafter [22], ETVA [23], and DEVIL [24] decompose the evaluation into a number of distinct dimensions. These typically include visual fidelity (per-frame quality), temporal consistency (e.g. smoothness, object permanence), motion plausibility (naturalness, physical adherence), content fidelity (semantic alignment with textual prompts), variety, dynamics (motion intensity and control), and increasingly, physical plausibility and common sense reasoning [21, 22]. These benchmarks employ diverse automated metrics, emphasizing alignment with human perception through direct user studies or by using large models as proxies for human judgment [21, 22].

Take Sora and Veo 2 as an example: By applying both VBench’s objective metrics and a frame-by-frame human review to the eight videos (Fig. 1) generated by Sora and Veo 2, we can directly observe how quantitative scores compare with human assessment results.

Quantitative evaluation with VBench: The application of VBench’s comprehensive assessment framework to the four experimental prompts (a-d) reveals distinct performance characteristics between Sora and Veo 2 (Fig. 2). The radar visualizations effectively represent multidimensional performance variations across essential metrics, with Sora demonstrating significant advantages in subject consistency, background consistency, and overall coherence for prompt (a). In contrast, Veo 2 exhibits enhanced capabilities in motion smoothness and achieves higher overall consistency scores for prompts (b) and (d). Besides, Veo 2 surpasses Sora in aesthetic quality and imaging fidelity for prompts (c) and (d), indicating superior capacity for fine detail reconstruction. However, Sora’s elevated temporal style score for prompt (a) suggests enhanced narrative coherence throughout the generated sequence.

These performance differences correspond, to some extent, to the fundamental architectural differences between

the models. Sora’s DiT framework utilizes Transformer architectures to model extensive contextual dependencies [14], enabling sophisticated temporal representation but occasionally manifesting structural inconsistencies across frames [3]. Conversely, the LDM approach of Veo 2 emphasizes computational efficiency through latent space compression [12], which enhances generation speed but potentially introduces reconstruction artifacts that affect prompt adherence. These aligned quantitative differences underscore the necessity for a VA-driven visual prescription workflow to diagnose and iteratively correct these failure modes [21, 25, 26]. We subsequently complement these findings with human perception results to reveal nuanced artifacts that objective metrics alone cannot adequately capture.

Human assessment results: Qualitative frame-by-frame analysis of four representative prompts (Fig. 1) reveals distinct deficiencies in both Sora and Veo 2 outputs. In prompt (a), Sora’s generation exhibits a positional inversion, interchanging the penguin’s head and tail during rotation. Conversely, Veo 2 produced a *Spheniscus humboldti* (a strange-skin endangered penguin) with an identification band (not present in the prompt but commonly found on this species of penguin) on its flipper and maintained consistent representation throughout the sequence, demonstrating the model’s imaginative intelligence. For prompt (b), Sora displays misaligned book pages during turning transitions, creating discontinuities in the visual flow. Meanwhile, Veo 2 renders textual content with mathematically inaccurate formulations and illustrations that fail to correspond to the specified prompt. In prompt (c), Sora’s butterfly’s wings appear as static entities while flying, with intermittent disappearance and spatial displacement between frames. The butterflies generated by Veo 2 are accompanied by a sudden color transition from purple to blue, disrupting temporal coherence. For prompt (d), Sora introduces unfounded footprint patterns on the desert surface and generates human figures with anatomically im-

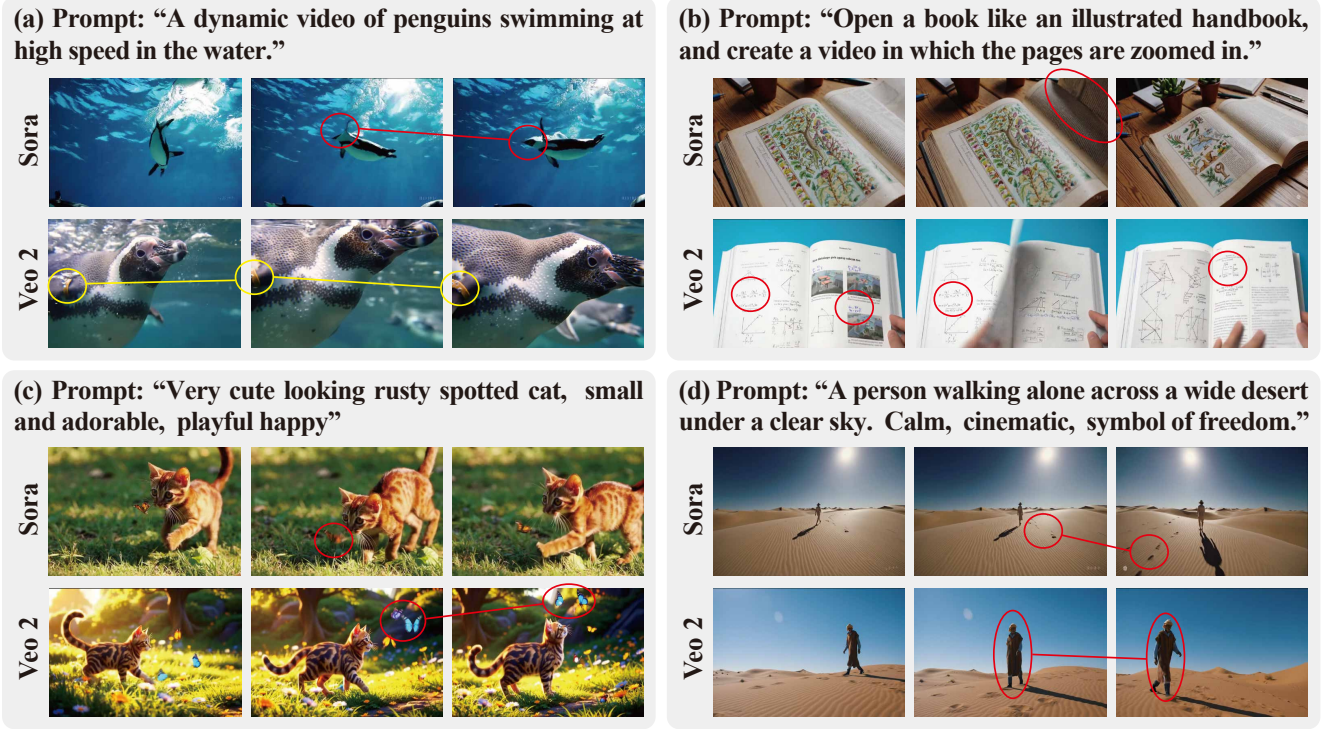


Figure 1: Screenshots of four prompts generated by Sora and Veo 2. We highlight inadequacies (e.g., discontinuous dynamics and phenomena that violate physical laws) by red circles and annotate a reasonable detail derived from imaginative intelligence with a yellow circle.

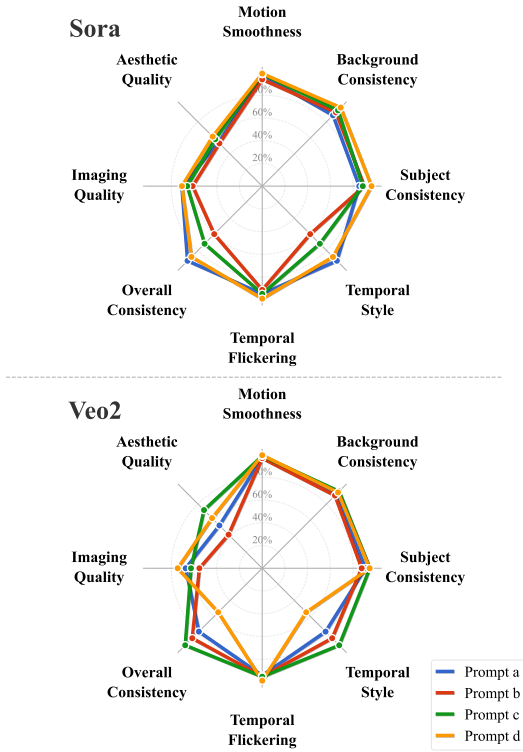


Figure 2: Radar charts display some key VBench metrics across the four example prompts (a-d) in Sora and Veo 2 through eight main dimension indicators.

plausible limb configurations and shadow projections inconsistent with bodily movements. Veo 2’s sequence is characterized by disjointed perspective shifts that compromise narrative continuity.

These nuanced perceptual anomalies, undetectable through automated evaluation protocols, suggest that automated assessment alone cannot accurately and unambiguously identify inadequacies, underscore the necessity for visual analytics-based diagnostic methodologies, which offer inherent interpretability advantages for identifying, analyzing, and potentially rectifying such multidimensional inadequacies.

2.3. Interactive Techniques from LLMs to MLLMs

From the above evaluation results, we can see that despite significant advancements in LLMs, persistent challenges exist in maintaining consistent factual accuracy and eliminating hallucinations, particularly when addressing complex tasks or edge cases (e.g., reference errors

in scientific writing [27], inconsistent proofs or calculations in advanced mathematical tests [28]). To address these limitations—and to investigate their applicability in MLLMs—researchers have developed two complementary categories of interactive techniques: feedback refinement mechanisms and interpretability tools [29].

Feedback refinement mechanisms: Contemporary human-AI collaboration extends far beyond basic prompt-based interactions, evolving toward more sophisticated feedback loops and interactive refinement mechanisms. To mitigate reliability issues concerning the inferential capabilities of LLMs, researchers are developing interactive engineering methodologies that facilitate iterative improvement. Analyzing workflow patterns analogous to software development, White et al. [30] identified multiple interaction patterns that enhance model performance through systematic feedback mechanisms, including output customization, error identification, and context control.

1) Output customization. Users can define the desired format of model outputs by specifying structural requirements within prompts. Beyond providing direct answers, techniques such as chain-of-thought prompts [31] direct LLMs to incorporate intermediate reasoning steps in their responses. Research demonstrates that chain-of-thought prompting yields improved accuracy in reasoning tasks. Furthermore, LLMs can synthesize multimodal information into their outputs.

2) Error identification. LLMs can detect errors through user-specified verification protocols. Responses to chain-of-thought prompts can undergo validation through deductive reasoning processes. The intermediate reasoning steps decompose complex questions into manageable sub-questions. This structured approach enables LLMs to implement self-verification through a divide-and-conquer methodology. Such an approach generates superior feedback by establishing transparent evaluation frameworks where errors can be localized to specific reasoning components, thus enabling targeted refinements.

3) Contextual control. LLMs can produce diverse answers to the same input. When seeking specific responses, users can eliminate irrelevant outputs by establishing contextual parameters through role-play prompts, intent-based prompts [30], and related techniques. For instance, a proficient LLM will employ accessible vocabulary when instructed to simulate a kindergarten teacher communicating with children. This contextual specification enhances feedback quality by constraining the model’s response domain to align with field-specific requirements, generating more relevant and actionable insights customized to particular applications or expertise levels.

Empirical evidence demonstrates that human users can elicit high-quality feedback from LLMs through methodically constructed prompts. Nevertheless, non-specialist users encounter significant difficulties when attempting to formulate or refine effective prompts [32].

Explainability and interpretability. To elucidate the internal mechanisms of LLMs, researchers have developed numerous analytical methods that illuminate decision processes and attribute importance to various features. *Feature visualization* (activation maximization) identifies patterns at the neuronal level [33]; saliency maps and integrated gradients quantify the relative importance of input tokens [34, 35]; while mechanistic circuit analysis isolates simplified subnetworks to delineate computational pathways [36, 37]. Complementary techniques, including hidden state classifiers [38], attention weight visualizations, and logit attribution [39] facilitate the localization of specific knowledge representations or reasoning patterns within the model architecture. Notwithstanding these methodological advances, contemporary analytical tools address only segments of an LLM’s extensive computational graph, face challenges related to scalability, and lack standardized metrics for evaluating the comprehensiveness of explanations [40, 41].

Extending to MLLMs. While MLLMs offer the potential for integrated processing of text, images, and video,

they introduce novel challenges in both prompt engineering and system interpretability. Multimodal prompts necessitate the encoding of heterogeneous input (e.g., analyze a diagram and synthesize key points), yet the literature reveals a paucity of frameworks to assist users in developing effective cross-modal instructions. Regarding interpretability, current methodologies such as saliency mapping and circuit tracing demonstrate insufficient capability to differentiate visual feature pathways or elucidate cross-modal attention mechanisms. Consequently, non-specialist users experience considerable difficulty when attempting to articulate visual parameters or identify the source of multimodal processing errors. Addressing these limitations will require innovative user interface architectures supporting iterative prompt refinement across different modalities, comprehensive multimodal provenance systems that document how individual inputs influence generated outputs, and integrated analytical approaches that combine MLLM interpretability with visual analytics techniques.

2.4. Requirements of Visual Prescription

To take full advantage of the imaginative intelligence of large Sora-like models, humans need assistance in blocking side effects through visual prescription. According to the progress of imaginative intelligence and related studies on LLMs, we propose the following requirements for visual prescription.

R1: Describe the authoring idea. This step illuminates the creative thought process behind it, revealing how the model understood the Prompt, what it chose to emphasize, what it added through its own imaginative Intelligence, and how its technical architecture shaped these decisions.

R2: Follow a user-specified fact-check list. In order to catch subtle or easily overlooked errors in the video, the system must generate comprehensive evaluation checklists. But we recognize that the universe of potential

facts is enormous, making complete verification impractical. So, we need users to guide the fact-checking process toward areas most relevant to their specific content and creative goals. This step leverages the strengths of both human knowledge and imaginative intelligence to create a fact-checking process that maximizes accuracy while optimizing resource utilization.

R3: Support efficient defect identification. Videos generated by current models exhibit diverse failure modes that remain undetectable through automated evaluation alone (e.g., Sora’s positional inversions or Veo 2’s discontinuous perspectives). The system should integrate visual analytics with automated detection mechanisms to rapidly highlight physical inconsistencies, semantic misalignments, and hallucinations across frames, substantially reducing review effort while increasing accuracy.

R4: Spark inspiration for optimization scheme. Drawing from the insights generated during evaluation, the system should actively engage users in interpretive dialogue, transforming detected issues into creative opportunities. By highlighting both flaws and latent potentials in the generated video, it invites users to reflect, iterate, and reimagine their narrative or aesthetic approach. This bidirectional feedback loop becomes a catalyst for innovation, enabling users to refine prompts, adjust model parameters, or recalibrate scene design with newfound clarity and purpose.

R5: Support precision editing for existing videos. For identified errors, the system should implement localized editing rather than complete regeneration, precisely addressing issues without introducing new uncertainties.

R6: Enable human and AI collaborative creation. The system should support real-time interaction between human intent and AI generation, allowing users to guide, adjust, and refine outputs iteratively. This tight feedback loop merges machine precision with human creativity to streamline complex video-authoring tasks.

R7: Brainstorm video authoring ideas with people. Beyond human-AI collaboration, the system should foster environments where human creators can ideate together, combining diverse perspectives, domain expertise, and narrative intuition. By facilitating shared workspaces, modular ideation tools, and concept visualizations, it enables collaborative exploration of storylines, visual motifs, and thematic arcs. Amplified by intelligent scaffolding, this human-to-human synergy becomes a foundation for richer, resonant video narratives.

3. Visual Prescriptions for Imaginative Intelligence

In this section, we propose a visual prescription workflow to meet the partial requirements of Section 2.4, and discuss how VA approaches can be integrated into the workflow to meet those requirements.

3.1. Workflow

As illustrated in Fig. 3, the authoring workflow operates through two interlinked components: imaginative intelligence and visual prescription. The visual prescription process begins with task-related prompts (R5), which initiate MLLM-based video authoring and generate an initial video draft along with a corresponding visual concept (R1). This concept serves as the entry point into an iterative loop of analysis and refinement. A fact checklist (R2) is constructed based on both domain-specific and general constraints, which is then evaluated through quality assessment (R3). Insights from R3 inform localized edits and optimization strategies (R4), feeding back into prompt refinement (R5) and potentially regenerating or adjusting the visual concept (R1). Rather than following a linear sequence, these stages operate in a dynamic cycle, with VA subtly integrated throughout to aid interpretability, traceability, and semantic alignment. Human-AI collaboration (R6) and multi-user coordination (R7) span across the workflow, enabling transparent tracking of provenance,

collaborative revisions, and consistent alignment with creative intent.

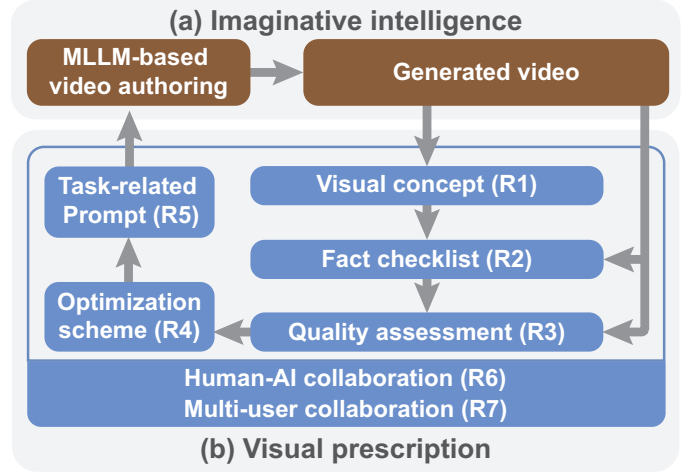


Figure 3: The workflow of visual prescription for imaginative intelligence.

3.2. Research Opportunities in VA

The proposed workflow reveals six critical research opportunities at the intersection of VA and generative MLLM, each addressing unmet requirements from the prescription paradigm:

O1: Multimodal semantic alignment framework.

While MLLMs demonstrate proficiency in natural language communication, they exhibit limitations in interpreting domain-specific terminology and subjective expressions, which impede precise human-AI alignment. The Visual Concept refers to a structured, multimodal representation that binds abstract textual prompts with concrete visual elements (e.g., images, keyframes, or short clips). This concept serves as the semantic anchor that resolves ambiguities arising from subjective or underspecified instructions (e.g., “cinematic ambiance”) [42], enabling consistent interpretation across stages and facilitating translation from conceptual specifications to actionable visual parameters, which subsequently inform the fact checklist (R2) validation process.

O2: Human-in-the-Loop error detection.

The fact checklist is a modular, logic-driven structure for en-

coding and validating content-specific factual constraints. Informed by the visual concept, this component empowers users to define critical error patterns (e.g., continuity lapses, misbehavior of certain characters) through interactive rule-building tools. It transforms the fact-checking task into a guided, explainable, and user-prioritized evaluation layer that supports high-precision corrections and quality benchmarking [43, 44].

O3: Multi-objective video evaluation. Generated video content undergoes a systematic verification against a fact checklist to ensure factual accuracy. Concurrently, the content is subjected to multidimensional assessment parameters that evaluate various quality aspects. This comprehensive analytical framework produces a structured evaluation report that documents both factual adherence and quality metrics. The report integrates with the optimization scheme (R4) to facilitate iterative refinement based on quantifiable performance indicators.

O4: Quality-resource efficiency trade-offs. This phase addresses tensions between video quality and computational constraints. It processes reports from quality assessment (O3) and prioritizes them based on severity, frequency, and resource impact. Rather than relying on full-scene regeneration, it recommends targeted prompt or parameter modifications for localized corrections [45].

O5: Integrated multimodal engineering. Text prompts alone often fall short in conveying complex user intentions [46]. The Task-related Prompt introduces a composite structure that fuses textual instructions with visual cues, spatial annotations, and semantic metadata. This multimodal configuration enables targeted operations, such as ensuring object continuity across scene transitions or regenerating environments without disturbing character posture. Significantly, it embeds optimization strategies (R4), balancing visual fidelity with system efficiency. By incorporating constraints directly into the prompt architecture, it aligns user intent with model execution. Techniques like contextual control (e.g., role-play

prompting) further refine output by narrowing interpretative variance, enhancing consistency and relevance in task-related prompts [30].

O6: Collaborative brainstorming. While O1 through O5 primarily address mechanisms of human-AI interaction (R6), effective video authoring at scale also demands structured collaboration among human creators. Multi-user collaboration refers to a synchronized, provenance-aware editing environment designed to support distributed, team-based creative workflows. This environment transparently captures the decision trajectory by integrating time-stamped annotations, iterative version lists, and parameter modification histories across modular visualization interfaces [47]. Aligned with R7, it extends beyond individual human-AI exchanges to enable real-time tracking of edits, team annotations, and conflict resolution across contributors. The result is a shift from linear content development to a networked, iterative design process rooted in transparency, accountability, and collaborative ideation.

4. A Use Case for Visual Prescription

This section discusses a possible practical scenario that demonstrates the effectiveness of the proposed visual prescription workflow. Using the video generated by Veo 2 based on prompt (d) in Fig. 2 as an example, we describe how iterative visual prescription can resolve anomalies (e.g., disjointed perspective and inconsistent shadow).

Following the visual prescription workflow, we extract the visual concept of a lone figure in traditional desert robes and a hat, expansive rippled dunes under a bright blue sky, the forward walking motion paired with a slow camera pan, and the overarching theme of a solitary desert journey. In addition to basic character dynamics and physics rules, perspective and shadows were marked as key fact rules by users. Character appearance, however, was excluded from the list. The VA tool facilitates us in identifying incoherent perspective shifts. We then follow

the fact-checking results and modify the prompt to the optimized prompt (revision of the prompt is colored blue in Fig. 4 (b)). As shown in the revised video, the camera angle and the effectiveness of shadow matching are improved.

The overall consistency and temporal style of the regenerated output are significantly improved (Fig. 4 (c)), and camera perspective changes and shadow mismatches are resolved. This demonstrates how visual prescriptions can improve cue accuracy and user coordination.

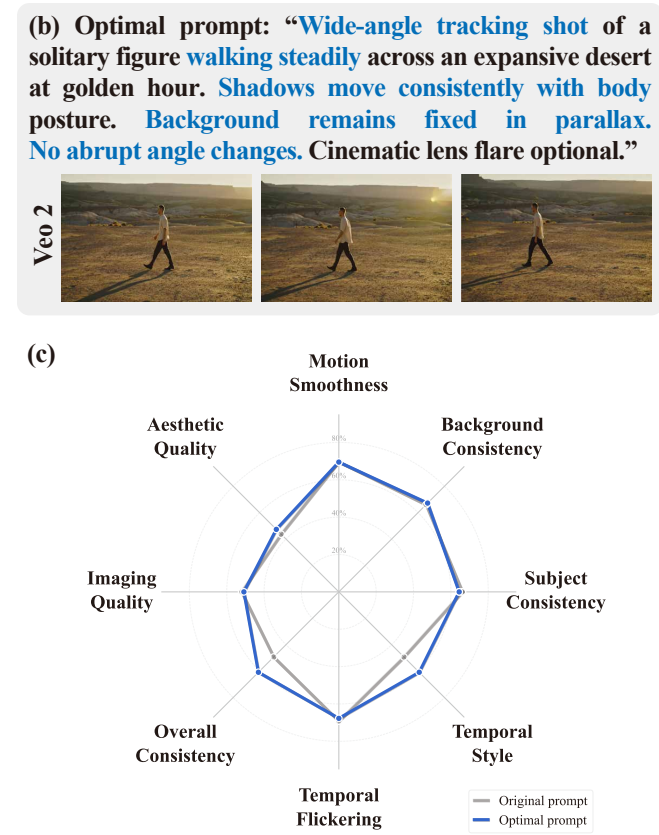


Figure 4: Video improved through a visual prescription process.

5. Conclusion

This work presents a comprehensive review of imaginative intelligence in MLLMs, proposes a visual prescription framework powered by visual analytics, and identifies six research opportunities at the intersection of generative AI and VA. The framework systematically addresses hallucination, semantic misalignment, and interpretability chal-

lenges through human-in-the-loop correction, multimodal prompt refinement, and collaborative creativity support. By aligning prescription requirements with VA opportunities, it provides a structured methodology for improving the controllability and transparency of MLLM-based video generation. While primarily conceptual, the framework offers a practical foundation for building trustworthy, efficient, and co-creative systems. Future work will focus on empirical validation, integration with explainable AI, and interdisciplinary system design.

Declaration of Generative AI and AI-assisted technologies in the writing process

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Claude3.7 Sonnet only to improve readability and language quality. After using this tool, the author(s) reviewed and edited the content as needed and took full responsibility for the content of the publication.

References

- [1] J. Wang, H. Jiang, Y. Liu, et al., A Comprehensive Review of Multimodal Large Language Models: Performance and Challenges Across Different Tasks, arXiv preprint arXiv:2408.01319 (2024). doi:10.48550/arXiv.2408.01319.
- [2] D. Caffagni, F. Cocchi, L. Barsellotti, et al., The Revolution of Multimodal Large Language Models: A Survey, arXiv preprint arXiv:2402.12451 (2024). doi:10.48550/arXiv.2402.12451.
- [3] OpenAI, Video generation models as world simulators (2024). URL <https://openai.com/research/video-generation-models-as-world-simulators>
- [4] Google DeepMind, State-of-the-art video and image generation with Veo 2 and Imagen 3 (Dec 2024). URL <https://blog.google/technology/google-labs/video-image-generation-update-december-2024/>

- [5] e. a. Bar-Tal, Lumiere: A Space-Time Diffusion Model for Video Generation, in: SIGGRAPH Asia 2024 Conference Papers, SA '24, Association for Computing Machinery, New York, NY, USA, 2024. doi:10.1145/3680528.3687614.
- [6] W. Wang, W. Lei, F. Ma, G. Huang, L. Liu, A Comprehensive Survey on Human Video Generation: Challenges, Methods, and Insights, arXiv preprint arXiv:2407.08428 (2024). doi:10.48550/arXiv.2407.08428.
- [7] Y. Zhang, Y. Li, L. Cui, D. Cai, L. Liu, Y. Fu, Hallucination of Multimodal Large Language Models: A Survey, arXiv preprint arXiv:2404.18930 (2024). doi:10.48550/arXiv.2404.18930.
- [8] Y. Wang, Q. Liu, C. Jin, Is RLHF More Difficult than Standard RL? A Theoretical Perspective, in: Advances in Neural Information Processing Systems, Vol. 36, Curran Associates, Inc., 2023, pp. 76006–76032.
- [9] J. Li, W. Gan, Z. Chen, et al., Multimodal Large Language Models: A Survey, arXiv preprint arXiv:2311.13165 (2024). doi:10.48550/arXiv.2311.13165.
- [10] W. Lei, J. Wang, F. Ma, G. Huang, L. Liu, A Comprehensive Survey on Human Video Generation: Challenges, Methods, and Insights, arXiv preprint arXiv:2407.08428 (2024). doi:10.48550/arXiv.2407.08428.
- [11] W. Peebles, S. Xie, Scalable diffusion models with transformers, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 4195–4205. doi:10.48550/arXiv.2212.09748.
- [12] R. Rombach, A. Blattmann, et al., High-resolution image synthesis with latent diffusion models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 10684–10695. doi:10.48550/arXiv.2112.10752.
- [13] H. Chen, C. Li, X. Tan, Y. Gao, L. Mai, Y. Liu, A Survey on Generative Diffusion Models, IEEE Transactions on Knowledge and Data Engineering (2024). doi:10.1109/TKDE.2024.3361474.
- [14] A. Vaswani, N. Shazeer, et al., Attention is All you Need, in: I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, R. Garnett (Eds.), Advances in Neural Information Processing Systems, Vol. 30, Curran Associates, Inc., 2017.
- [15] Z. Li, Y. Zhang, T. Chen, et al., LatentExplainer: Explaining Latent Representations in Deep Generative Models with Multimodal Foundation Models, arXiv preprint arXiv:2406.14862 (2024). doi:10.48550/arXiv.2406.14862.
- [16] D. Keim, G. Andrienko, et al., Visual analytics: Definition, process, and challenges, in: Information visualization: Human-centered issues and perspectives, Springer, 2008, pp. 154–175. doi:10.1007/978-3-540-70956-5_7.
- [17] K. A. Cook, J. J. Thomas, Illuminating the Path: The Research and Development Agenda for Visual Analytics, IEEE Computer Society, Los Alamitos, CA, USA, 2005.
- [18] Z. Wang, A. C. Bovik, H. R. Sheikh, E. P. Simoncelli, Image quality assessment: from error visibility to structural similarity, IEEE Transactions on Image Processing 13 (4) (2004) 600–612. doi:10.1109/TIP.2003.819861.
- [19] T. Unterthiner, et al., Towards Accurate Generative Models of Video: A New Metric & Challenges, arXiv preprint arXiv:1812.01717 (2018). doi:10.48550/arXiv.1812.01717.
- [20] J. Carreira, A. Zisserman, Quo Vadis, Action recognition? A New Model and the Kinetics Dataset, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017. doi:10.48550/arXiv.1705.07750.
- [21] Z. Huang, Y. He, J. Yu, et al., VBench: Comprehensive Benchmark Suite for Video Generative Models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024. doi:10.48550/arXiv.2311.17982.
- [22] Y. Liu, X. Cun, et al., Evalcrafter: Benchmarking and Evaluating Large Video Generation Models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 22139–22149. doi:10.48550/arXiv.2310.11440.
- [23] K. Guan, Z. Lai, et al., ETVA: Evaluation of Text-to-Video Alignment via Fine-grained Question Generation and Answering, arXiv preprint arXiv:2503.16867 (2025). doi:10.48550/arXiv.2503.16867.
- [24] R. Szeto, J. J. Corso, The DEVIL Is in the Details: A Diagnostic Evaluation Benchmark for Video Inpainting, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 21054–21063. doi:10.48550/arXiv.2105.05332.
- [25] Z. Huang, F. Zhang, X. Xu, et al., VBench++: Comprehensive and Versatile Benchmark Suite for Video Generative Models, arXiv preprint arXiv:2411.13503 (2024). doi:10.48550/arXiv.2411.13503.
- [26] D. Zheng, Z. Huang, H. Liu, et al., VBench-2.0: Advancing Video Generation Benchmark Suite for Intrinsic Faithfulness, arXiv preprint arXiv:2503.21755 (2025). doi:10.48550/arXiv.2503.21755.
- [27] H. Alkaissi, S. I. McFarlane, Artificial Hallucinations in ChatGPT: Implications in Scientific Writing, Cureus 15 (2) (2023). doi:10.7759/cureus.35179.
- [28] S. Frieder, L. Pinchetti, R.-R. Griffiths, et al., Mathematical Capabilities of ChatGPT, Advances in Neural Information Processing Systems 36 (2024). doi:10.48550/arXiv.2301.13867.
- [29] Y. Dang, K. Huang, et al., Explainable and Interpretable Multimodal Large Language Models: A Comprehensive Survey, arXiv

- preprint arXiv:2412.02104 (2024). doi:10.48550/arXiv.2412.02104.
- [30] J. White, Q. Fu, et al., A Prompt Pattern Catalog to Enhance Prompt Engineering with Large Language Models, arXiv preprint arXiv:2302.11382 (2023). doi:10.48550/arXiv.2302.11382.
- [31] J. Wei, X. Wang, D. Schuurmans, et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, in: Proceedings of the 36th International Conference on Neural Information Processing Systems, Curran Associates Inc., 2022. doi:10.5555/3600270.3602070.
- [32] J. Zamfirescu-Pereira, R. Y. Wong, B. Hartmann, Q. Yang, Why Johnny Can’t Prompt: How Non-AI Experts Try (and Fail) to Design LLM Prompts (2023) 1–21doi:10.1145/3544548.3581388.
- [33] C. Olah, A. Mordvintsev, et al., Feature Visualization, Distill (2017). doi:10.23915/distill.00007.
- [34] M. Sundararajan, A. Taly, Q. Yan, Axiomatic Attribution for Deep Networks, in: Proceedings of the 34th International Conference on Machine Learning, Vol. 70, PMLR, 2017, pp. 3319–3328.
- [35] R. Saleem, B. Yuan, F. Kurugollu, A. Anjum, L. Liu, Explaining Deep Neural Networks: A survey on the Global Interpretation Methods, Neurocomputing 513 (2022) 165–180. doi:10.1016/j.neucom.2022.09.129.
- [36] E. Ameisen, J. Lindsey, et al., Circuit Tracing: Revealing Computational Graphs in Language Models (2025).
URL <https://transformer-circuits.pub/2025/attribution-graphs/methods.html>
- [37] e. a. Lindsey, On the Biology of a Large Language Model (2025).
URL <https://transformer-circuits.pub/2025/attribution-graphs/biology.html>
- [38] G. Alain, Y. Bengio, Understanding Intermediate Layers Using Linear Classifiers, arXiv preprint arXiv:1612.00837 (2016). doi:10.48550/arXiv.1610.01644.
- [39] J. Vig, Y. Belinkov, Analyzing the Structure of Attention in a Transformer Language Model, in: Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP, 2019, pp. 63–76. doi:10.48550/arXiv.1906.04284.
- [40] Z. C. Lipton, The Mythos of Model Interpretability, ACM Queue 16 (3) (2018) 31–57. doi:10.1145/3236386.3241340.
- [41] Z. Wu, L. Zhu, K. Lin, et al., A Survey on Mechanistic Interpretability for Multi-Modal Foundation Models, arXiv preprint arXiv:2502.17516 (2025). doi:10.48550/arXiv.2502.17516.
- [42] A. Radford, J. W. Kim, et al., Learning Transferable Visual Models From Natural Language Supervision, in: Proceedings of the 38th International Conference on Machine Learning, Vol. 139, PMLR, 2021, pp. 8748–8763.
- [43] L. Pan, X. Lu, et al., QACHECK: A Demonstration System for Question-Guided Multi-Hop Fact-Checking, arXiv preprint arXiv:2310.07609 (2023). doi:10.48550/arXiv.2310.07609.
- [44] X. Ye, X. Chen, Z. Zhang, et al., Human-in-the-loop machine learning: a state of the art, Artificial Intelligence Review 55 (2022) 2971–3008. doi:10.1007/s10462-022-10246-w.
- [45] S. Liu, Y. Zhang, W. Li, Z. Lin, J. Jia, Video-P2P: Video Editing with Cross-attention Control, arXiv preprint arXiv:2303.04761 (2023). doi:10.48550/ARXIV.2303.04761.
- [46] S. Yin, C. Fu, S. Zhao, K. Li, X. Sun, T. Xu, E. Chen, A Survey on Multimodal Large Language Models, National Science Review 11 (2024) nwae403. doi:10.1093/nsr/nwae403.
- [47] D. C.-E. Lin, F. C. Heilbron, et al., VideoMap: Supporting Video Editing Exploration, Brainstorming, and Prototyping in the Latent Space, arXiv preprint arXiv:2211.12492 (2022). doi:10.48550/arXiv.2211.12492.